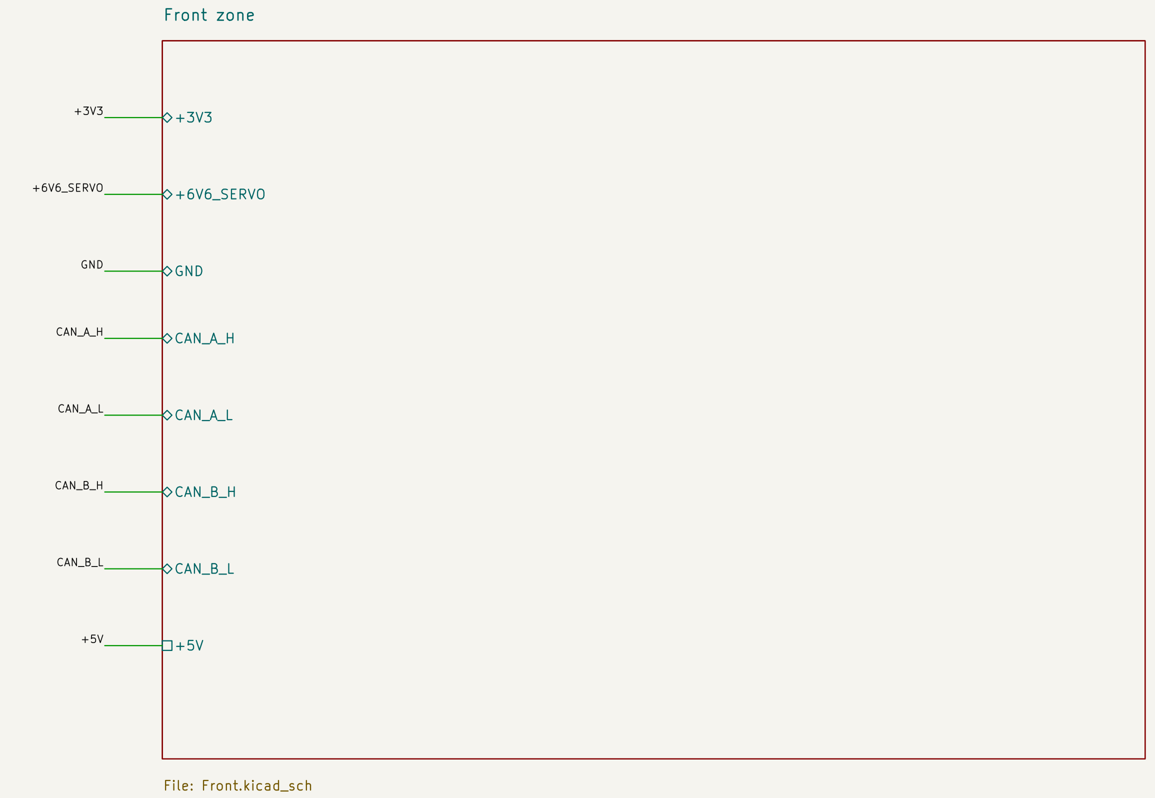
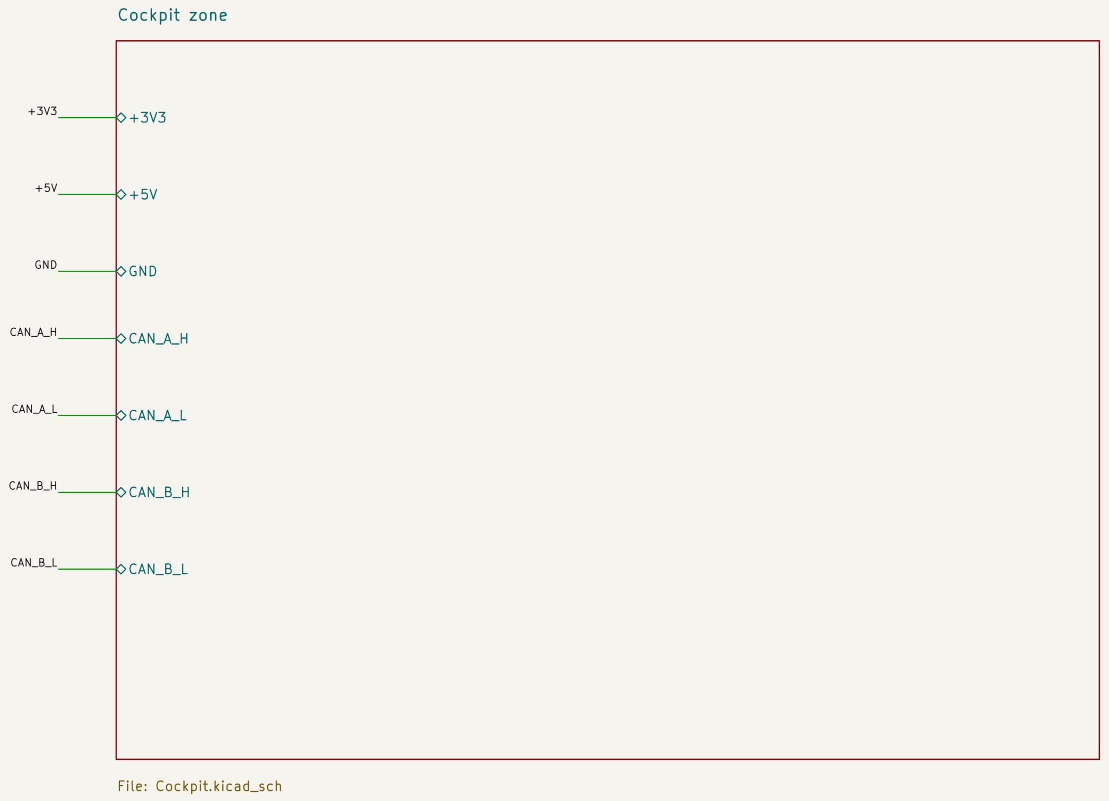
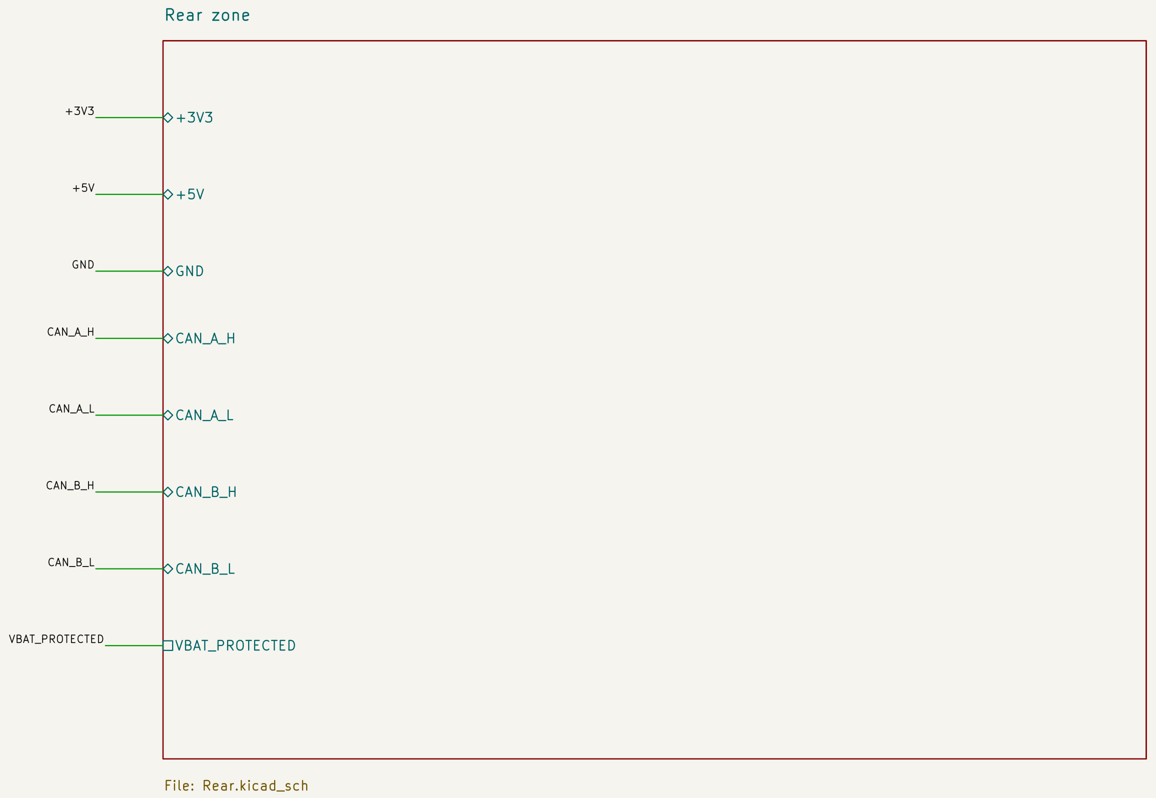
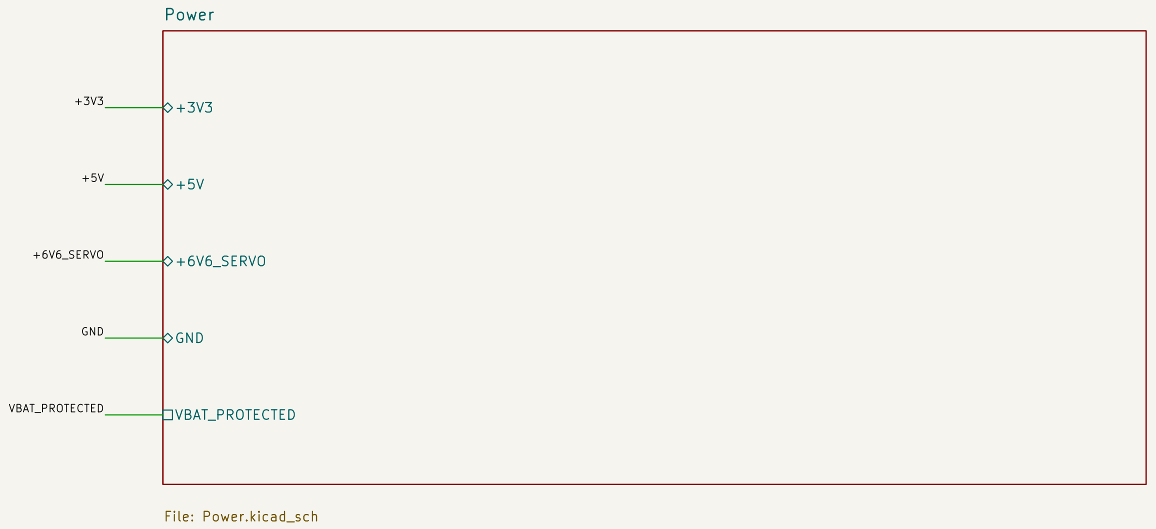
LAB 4 – CONNECTIVITY OVERVIEW

Front and rear are the terminated ends. Cockpit taps both backbones.
Each zone has its own two isolation groups; termination remains on the bus side.
Actuator connectors / motor protection / physical layout remain unfinished.


Front zone
+3V3
+6V6_SERVO
GND
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L
+5V
+3V3
+6V6_SERVO
GND
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L
+5V
File: Front.kicad_sch


Cockpit zone
+3V3
+5V
GND
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L
+3V3
+5V
GND
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L
File: Cockpit.kicad_sch


Rear zone
+3V3
+5V
GND
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L
VBAT_PROTECTED
+3V3
+5V
GND
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L
VBAT_PROTECTED
File: Rear.kicad_sch


Power
+3V3
+5V
+6V6_SERVO
GND
VBAT_PROTECTED
+3V3
+5V
+6V6_SERVO
GND
VBAT_PROTECTED
File: Power.kicad_sch

CAN_A_H/L and CAN_B_H/L are separate root nets. Same-name local labels on this root join the shown sheet pins. Child labels connect only through those ports.

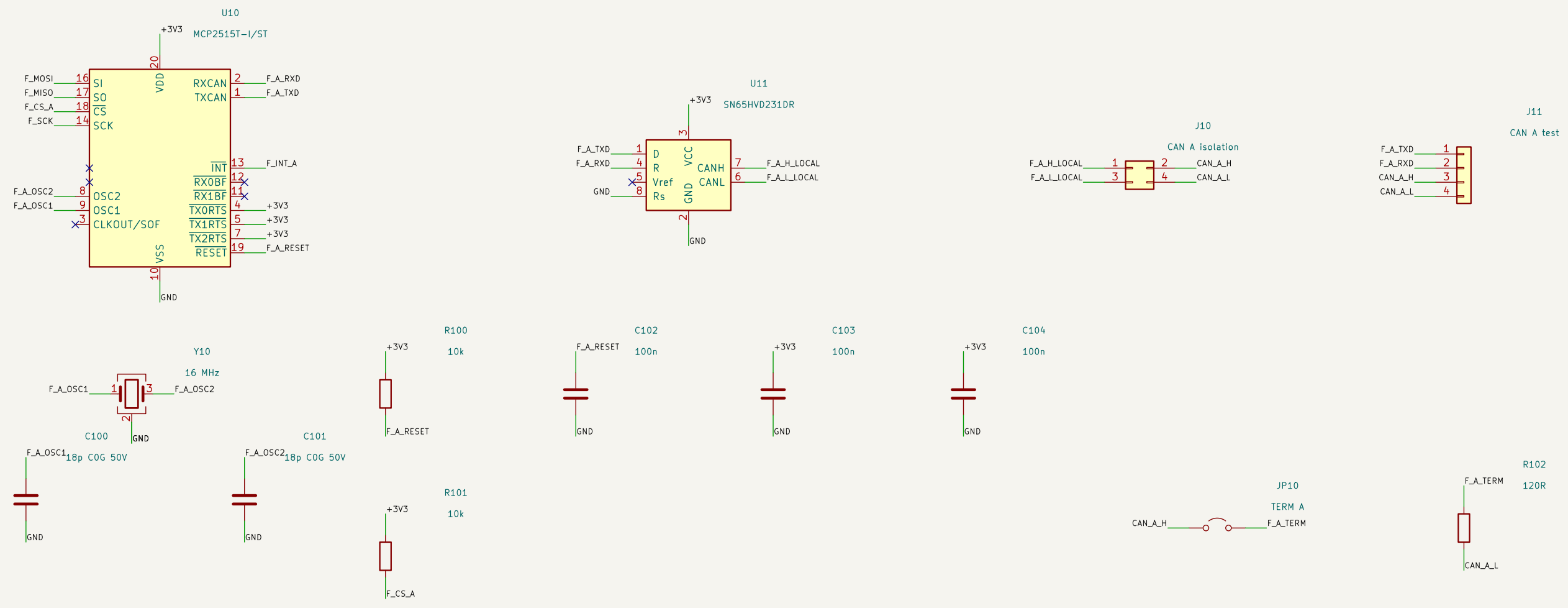
Power flags document external battery power and outputs after passive filters; they do not prove current capacity, regulation, thermal behavior or PCB completeness.

See Verification.txt and Design-review.csv for current checks and open decisions.

Shared nets: two CAN backbones, power and ground only.
PRELIMINARY – incomplete board; not for fabrication
Team 4
Sheet: /
File: Lab4.kicad_sch
Title: Lab 4 – three-zone controller
Size: A2 Date: 2026-10-20 **Rev: 0.3**
KiCad E.D.A. 10.0.4 Id: 1/5

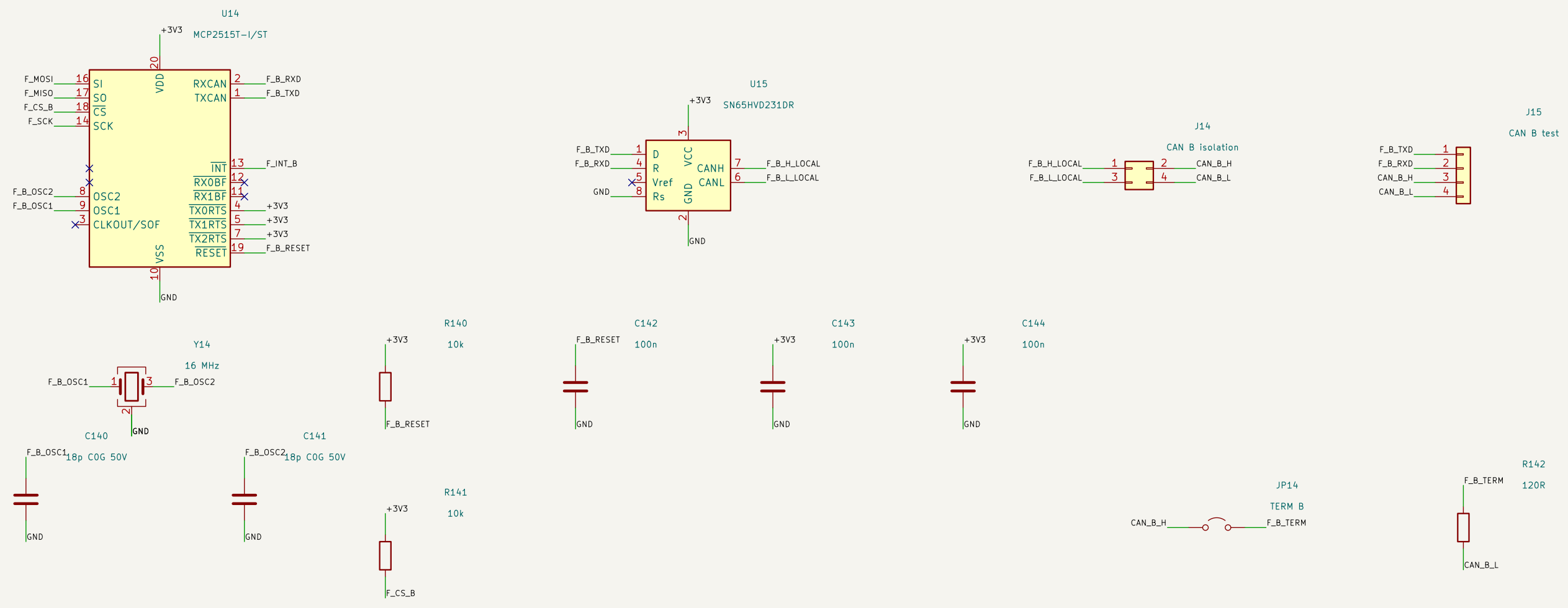
Front zone — CAN and servo current sensing

CAN A: 16 MHz / 500 kbit/s / 3.3 V


U10
MCP2515T-I/ST
U11
SN65HVD231DR
J10
CAN A isolation
J11
CAN A test
JP10
TERM A

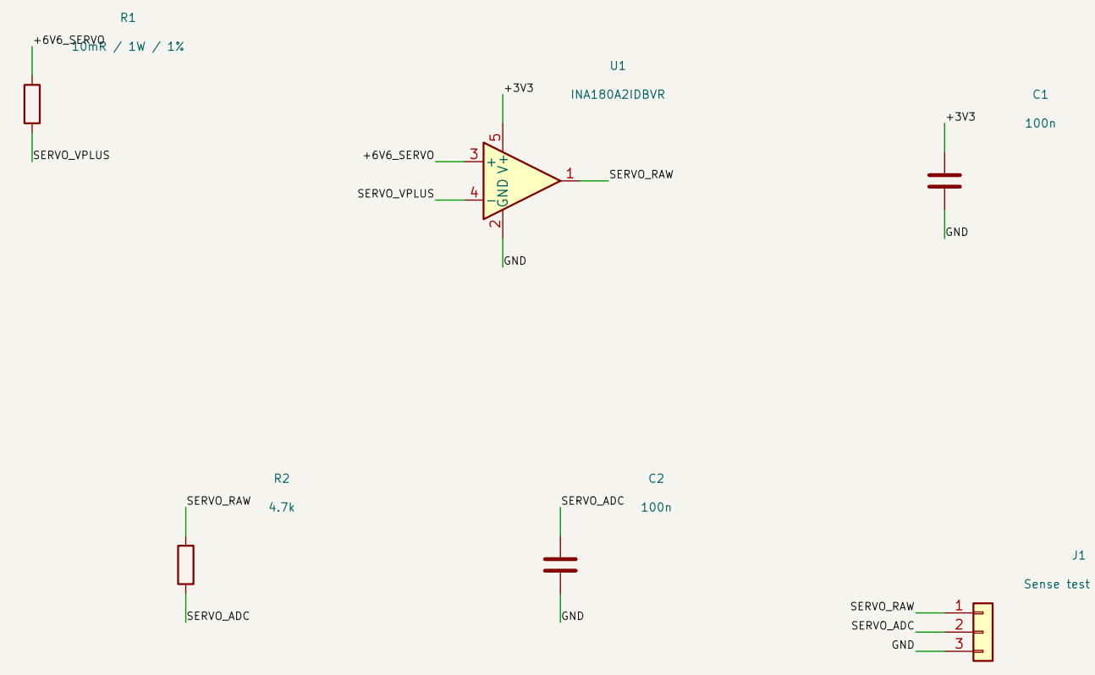
RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

CAN B: 16 MHz / 500 kbit/s / 3.3 V

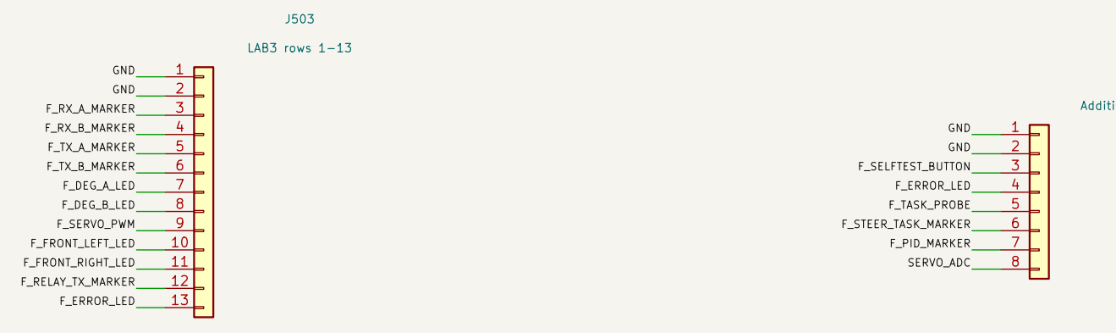

MCP2515T-I/ST
U15
SN65HVD231DR
J14
CAN B isolation
CAN B test

RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

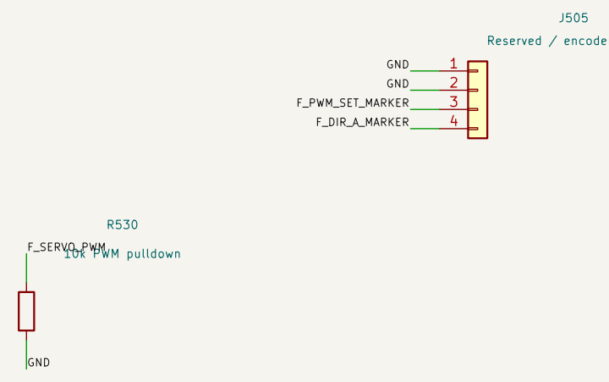
SERVO SHUNT — calculated proposal (not measured)


R1
10mR / 1W / 1%
U1
INA180A2IDBVR
+6V6_SERVO
SERVO_VPLUS
SERVO_RAW
SERVO_ADC
GND
Servo test points

6.6 V regulator -> 10 mOhm shunt -> servo +
Servo GND returns directly to power ground.
Kelvin sense: separate thin traces from shunt pads to IN+/IN-.
Gain 50: Vout = 0.5 V/A. Proposed range 0-5 A -> 0-2.5 V.
12-bit / 3.3 V: 1.61 mA/count (ideal quantization).
At 5 A: shunt drop 50 mV; loss 0.25 W. Select >=1 W.
RC: 4.7k + 100n -> 339 Hz; proposed ADC 4 ksample/s.
Near-zero amplifier output clips; calibrate offset and gain.
This unidirectional circuit cannot measure regenerated current.
No servo idle/stall/noise measurements yet.
5 A measurement range is NOT a 5 A driver or regulator rating.

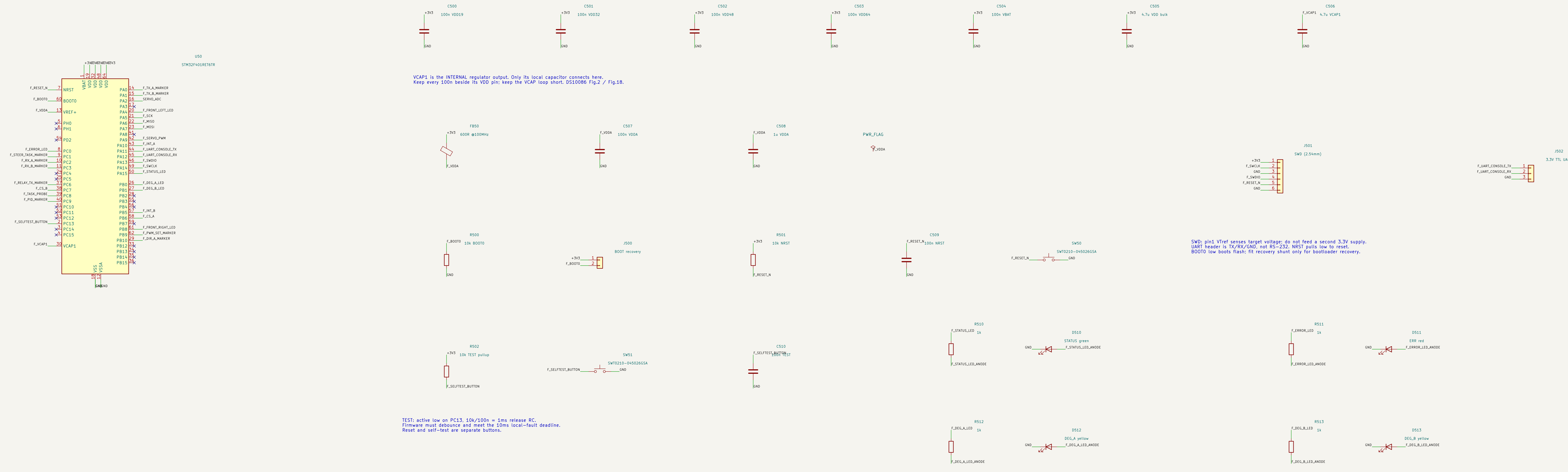

LAB3 rows 1-13
Additional probes
F_DEG_B_LED
F_SERVO_PWM
F_FRONT_LEFT_LED
F_FRONT_RIGHT_LED
F_RELAY_TX_MARKER
F_ERROR_LED

LAB3 header follows handout p17 top-to-bottom. Two GND pins first.
SRV/PWM_OUT probe actual actuator PWM; timing markers remain separate.
Physical header position / accessibility will be checked during PCB layout.


J505
Reserved / encoder probes
GND
GND
F_PWM_SET_MARKER
F_DIR_A_MARKER
R530
F_SERVO_PWM
10k PWM pulldown
GND

Servo and lamp circuits are below. Connector cable adapters still need physical verification.

STM32F401RET6 — independent HSI/PLL, no ST-LINK clock dependency

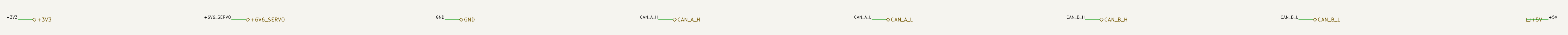

C500
100n VDD19
C501
100n VDD32
U50
STM32F401RET6TR
FB50
600R @100MHz
C507
100n VDDA
J501
SWD (2.54mm)
J502
3.3V TTL UART
R502
10k TEST pullup
R511
1k
D511
ERR red

VCAP1 is the INTERNAL regulator output. Only its local capacitor connects here.
Keep every 100n beside its VDD pin; keep the VCAP loop short. DS10086 Fig.2 / Fig.18.

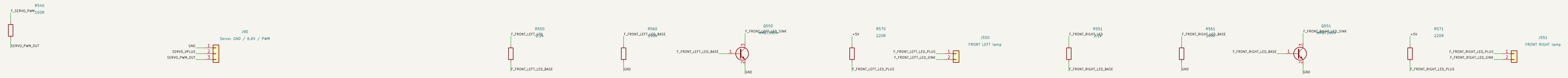
SWD: pin1 VTref senses target voltage; do not feed a second 3.3V supply.
UART header is TX/RX/GND, not RS-232. NRST pulls low to reset.
BOOT0 low boots flash; fit recovery shunt only for bootloader recovery.

TEST: active low on PC13, 10k/100n = 1ms release RC.
Firmware must debounce and meet the 10ms local-fault deadline.
Reset and self-test are separate buttons.

Shared rails and CAN backbone connect through the root sheet. SPI, clocks and reset stay within each zone.


+3V3
+6V6_SERVO
GND
CAN_A_H
CAN_A_L

SERVO AND VEHICLE LAMP OUTPUTS


R540
100R
F_SERVO_PWM
SERVO_PWM_OUT
J90
Servo: GND / 6.6V / PWM
GND
SERVO_VPLUS
SERVO_PWM_OUT
FRONT LEFT lamp
FRONT RIGHT lamp

Servo supply passes through R1. PWM uses PA9 and retains its reset pulldown.
J90 uses a new rated 3.96mm connector, not the existing servo plug.
Adapter must preserve GND / supply / signal; confirm the servo accepts 3.3V PWM.

Engineering review; actuator/protection and mechanical sign-off pending.
PRELIMINARY — incomplete board; not for fabrication
Team 4
Sheet: /Front zone/
File: Front.kicad_sch
Title: Front zone — integrated logic and interfaces
Size: A0 | Date: 2026-10-23 | **Rev: 0.3**
KiCad E.D.A. 10.0.4 | Id: 2/5

Rear zone – CAN and motor current sensing

CAN A: 16 MHz / 500 kbit/s / 3.3 V

RS = GND: high–speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

CAN B: 16 MHz / 500 kbit/s / 3.3 V

RS = GND: high–speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

LEFT MOTOR: L298 output –> ACS712 –> motor lead

RIGHT MOTOR: L298 output –> ACS712 –> motor lead

LAB3 header follows handout p17 top–to–bottom. Two GND pins first.
SRV/PWM_OUT probe actual actuator PWM; timing markers remain separate.
Physical header position / accessibility will be checked during PCB layout.

L298 and output circuits are below. Current limiting and heatsink remain unresolved; no motor–power sign–off.

Each motor: +/–5 A; VADC = 1.25 V + 0.0925 V/A * I.
Ideal 12–bit scale: 8.71 mA/count at VREF=3.3 V.
10k/10k divider keeps a 5 V sensor output at 2.5 V.
Filter source resistance: (10k || 10k) + 1k = 6k.
6k * 100n –> 265 Hz. Proposed ADC: 4 ksample/s/channel.
Stop motors before zero calibration; preserve signed readings.
Hall sensor has internal conductor, not an external shunt.
At 5 A and typical 1.2 mOhm, conductor loss = 30 mW.
Motor return lead connects to the OTHER L298 output, not GND.

STM32F401RET6 – independent HSI/PLL, no ST–LINK clock dependency

VCAP1 is the INTERNAL regulator output. Only its local capacitor connects here.
Keep every 100n beside its VDD pin; keep the VCAP loop short. DS10086 Fig.2 / Fig.18.

SWD: pin1 VTref senses target voltage; do not feed a second 3.3V supply.
UART header is TX/RX/GND, not RS–232. NRST pulls low to reset.
BOOT0 low boots flash; fit recovery shunt only for bootloader recovery.

TEST: active low on PC13, 10k/100n = 1ms release RC.
Firmware must debounce and meet the 10ms local–fault deadline.
Reset and self–test are separate buttons.

Shared rails and CAN backbone connect through the root sheet. SPI, clocks and reset stay within each zone.

ENCODER INPUTS: board pins 1=5.1V, 2=GND, 3=A, 4=B. Verify actual cable order.
10k pullups support open–collector outputs; push–pull input must stay <=5.5V.
U81 at 3.3V gives Schmitt hysteresis and MCU–safe outputs; unused inputs grounded.
Both A and B invert. Check direction/counts after the PCB firmware port.

MOTOR POWER STAGE – current–limit / heatsink sign–off still required

OUT1/OUT3 –> ACS712 IP+ –> IP– –> motor lead. Other lead –> OUT2/OUT4, NEVER ground.
ENA/ENB high + equal direction inputs gives dynamic braking. Enable low gives coast.
Existing 10k pulldowns keep enables low during reset. SENSE_A/B return to power ground.
This schematic has no autonomous motor–current limiter. A fuse and filtered ADC are not substitutes.
Use an agreed current–limit/driver redesign before motor power is enabled; 3.2A stall exceeds 2A.
The eight diodes clamp each bridge output to protected battery and ground. Keep loops short.

Engineering review; actuator/protection and mechanical sign–off pending.
PRELIMINARY – incomplete board; not for fabrication
Team 4
Sheet: /Rear zone/
File: Rear.kicad_sch
Title: Rear zone – integrated logic and interfaces
Size: A0
Date: 2026–10–23
Rev: 0.3
KiCad E.D.A. 10.0.4
Id: 3/5

Cockpit zone — dual CAN

RASPBERRY PI 4 — 40-pin board socket

CAN A: 16 MHz / 500 kbit/s / 3.3 V
U30
MCP2515T-I/ST
+3V3
PI_MOSI 16 SI
PI_MISO 17 SO
PI_CS_A 18 CS
PI_SCK 14 SCK
8 OSC2 PI_A_OSC2
9 OSC1 PI_A_OSC1
3 CLKOUT/SOF
20 VDD
10 VSS
GND
RXCAN 2 PI_A_RXD
TXCAN 1 PI_A_TXD
INT 13 PI_INT_A
RX0BF 12
RX1BF 11
TX0RTS 4 +3V3
TX1RTS 5 +3V3
TX2RTS 7 +3V3
RESET 19 PI_A_RESET
Y30
16 MHz
PI_A_OSC1
PI_A_OSC2
C300
18p C0G 50V
C301
18p C0G 50V
R300
10k
PI_A_RESET
R301
10k
PI_CS_A
U31
SN65HVD231DR
PI_A_TXD 1 D
PI_A_RXD 4 R
5 Vref
GND 8 Rs
3 VCC
2 GND
CANH 7 PI_A_H_LOCAL
CANL 6 PI_A_L_LOCAL
C302
100n
C303
100n
C304
100n
J30
CAN A isolation
PI_A_H_LOCAL 1 2 CAN_A_H
PI_A_L_LOCAL 3 4 CAN_A_L
J31
CAN A test
PI_A_TXD 1
PI_A_RXD 2
CAN_A_H 3
CAN_A_L 4
CAN B: 16 MHz / 500 kbit/s / 3.3 V
U34
MCP2515T-I/ST
PI_MOSI 16 SI
PI_MISO 17 SO
PI_CS_B 18 CS
PI_SCK 14 SCK
PI_B_OSC2 8 OSC2
PI_B_OSC1 9 OSC1
3 CLKOUT/SOF
RXCAN 2 PI_B_RXD
TXCAN 1 PI_B_TXD
INT 13 PI_INT_B
RX0BF 12
RX1BF 11
TX0RTS 4 +3V3
TX1RTS 5 +3V3
TX2RTS 7 +3V3
RESET 19 PI_B_RESET
Y34
16 MHz
PI_B_OSC1
PI_B_OSC2
C340
18p C0G 50V
C341
18p C0G 50V
R340
10k
PI_B_RESET
R341
10k
PI_CS_B
U35
SN65HVD231DR
PI_B_TXD 1 D
PI_B_RXD 4 R
5 Vref
GND 8 Rs
CANH 7 PI_B_H_LOCAL
CANL 6 PI_B_L_LOCAL
C342
100n
C343
100n
C344
100n
J34
CAN B isolation
PI_B_H_LOCAL 1 2 CAN_B_H
PI_B_L_LOCAL 3 4 CAN_B_L
J35
CAN B test
PI_B_TXD 1
PI_B_RXD 2
CAN_B_H 3
CAN_B_L 4
J70
Raspberry Pi4 GPIO
PI_ERROR_LED 32 GPIO12/PWM0
PI_STATUS_LED 33 GPIO13/PWM1
8 GPIO14/UART_TXD
10 GPIO15/UART_RXD
PI_SELFTEST_BUTTON 36 GPIO16/SPI1_CE2
PI_UDP_RX_MARKER 11 GPIO17/SPI1_CE1
12 GPIO18/SPI1_CE0/PCM_CLK/PWM0
35 GPIO19/SPI1_MISO/PCM_FS
38 GPIO20/SPI1_MOSI/PCM_DIN/PWM1
40 GPIO21/SPI1_SCLK/PCM_DOUT
PI_TX_A_MARKER 15 GPIO22/SDIO_CLK
PI_TX_B_MARKER 16 GPIO23/SDIO_CMD
PI_INT_B 18 GPIO24/SDIO_DAT0
PI_INT_A 22 GPIO25/SDIO_DAT1
37 GPIO26/SDIO_DAT2
PI_CMD_TX_MARKER 13 GPIO27/SDIO_DAT3
5V 2 +5V
3V3 1
ID_SD_I2C0/GPIO00 27
ID_SC_I2C0/GPIO01 28
SDA_I2C1/GPIO02 3
SCL_I2C1/GPIO03 5
GPCLK0/GPIO04 7
GPCLK1/GPIO05 29 PI_DEG_A_LED
GPCLK2/GPIO06 31 PI_DEG_B_LED
CE1_SPI0/GPIO07 26 PI_CS_B
CE0_SPI0/GPIO08 24 PI_CS_A
MISO_SPI0/GPIO09 21 PI_MISO
MOSI_SPI0/GPIO10 19 PI_MOSI
SCLK_SPI0/GPIO11 23 PI_SCK
GND 9
GND
J703
LAB3 rows 1–13
GND 1
GND 2
PI_UDP_RX_MARKER 3
PI_CMD_TX_MARKER 4
PI_TX_A_MARKER 5
PI_TX_B_MARKER 6
PI_DEG_A_LED 7
PI_DEG_B_LED 8
9
10
11
12
13
J704
Pi extra probes
GND 1
GND 2
PI_SELFTEST_BUTTON 3
PI_ERROR_LED 4
PI_STATUS_LED 5
C700
+5V
22u /16V
C701
+5V
100n
R700
+3V3
10k TEST pullup
PI_SELFTEST_BUTTON
SW70
SWT0210-045026GSA
PI_SELFTEST_BUTTON
GND
C702
PI_SELFTEST_BUTTON
100n TEST
R710
PI_STATUS_LED
1k
PI_STATUS_LED_ANODE
D710
STATUS green
GND
PI_STATUS_LED_ANODE
R711
PI_ERROR_LED
1k
PI_ERROR_LED_ANODE
D711
ERR red
GND
PI_ERROR_LED_ANODE
R712
PI_DEG_A_LED
1k
PI_DEG_A_LED_ANODE
D712
DEG_A yellow
GND
PI_DEG_A_LED_ANODE
R713
PI_DEG_B_LED
1k
PI_DEG_B_LED_ANODE
D713
DEG_B yellow
GND
PI_DEG_B_LED_ANODE
+3V3
+5V
GND
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L

RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

Pins 2/4 receive board 5.1V. Pi 3.3V pins 1/17 are NOT joined to board 3.3V.
Do not attach independent USB-C power while board 5.1V is connected.
Confirm socket orientation with a 1:1 print and the actual Pi before routing.

Rows 9–13 intentionally unused on Pi (Lab3 p17).
Two GND pins per probe group. Active-high LED outputs.
Physical reachability is a layout check, not established by this schematic.

Local bypass is not a load-step proof. Pi bulk capacitance / rail transient test remains open.

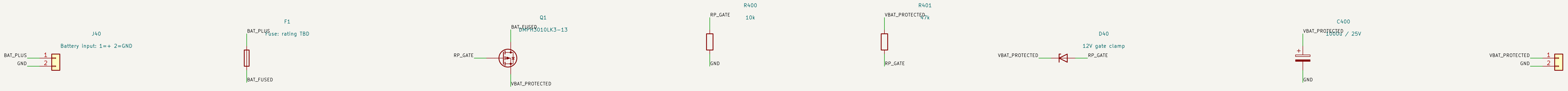
All Pi GPIOs are 3.3V signals. Header pin number is NOT BCM GPIO number.
Independent board 3.3V supplies the six CAN interfaces.
A=CE0/IRQ25; B=CE1/IRQ24; confirm Linux labels against the installed overlay.

Shared rails and CAN backbone connect through the root sheet. SPI, clocks and reset stay within each zone.

Engineering review; actuator/protection and mechanical sign–off pending.
PRELIMINARY – incomplete board; not for fabrication
Team 4
Sheet: /Cockpit zone/
File: Cockpit.kicad_sch
Title: Cockpit zone – integrated logic and interfaces
Size: A1
Date: 2026–10–23
Rev: 0.3
KiCad E.D.A. 10.0.4
Id: 4/5

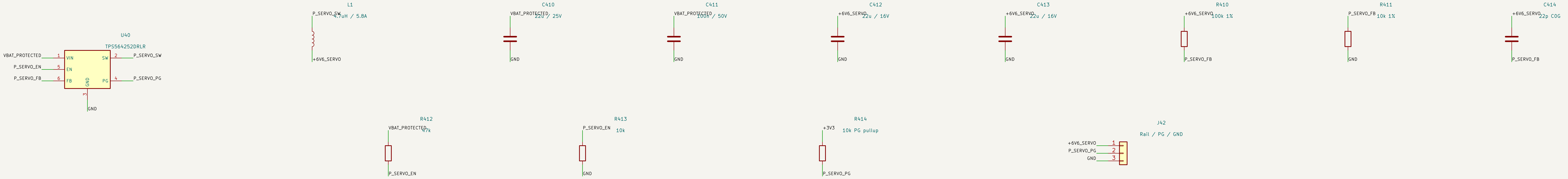
AA PACK INPUT — 10x NiMH recommended; exact cells/holder not selected

Design input envelope: 8–15 V at the protected node; verify real cells, contact resistance and sag.
Battery / motor rail is UNREGULATED. No lithium charger or lithium BMS is included.


J40
Battery input: 1=+ 2=GND
BAT_PLUS 1
GND 2
F1
BAT_PLUS
Fuse: rating TBD
BAT_FUSED
Q1
BAT_FUSED
DMPH3010LK3-13
RP_GATE
VBAT_PROTECTED
R400
RP_GATE
10k
GND
R401
VBAT_PROTECTED
47k
RP_GATE
D40
12V gate clamp
VBAT_PROTECTED
RP_GATE
C400
VBAT_PROTECTED
1000u / 25V
GND
J41
VBAT test
VBAT_PROTECTED 1
GND 2

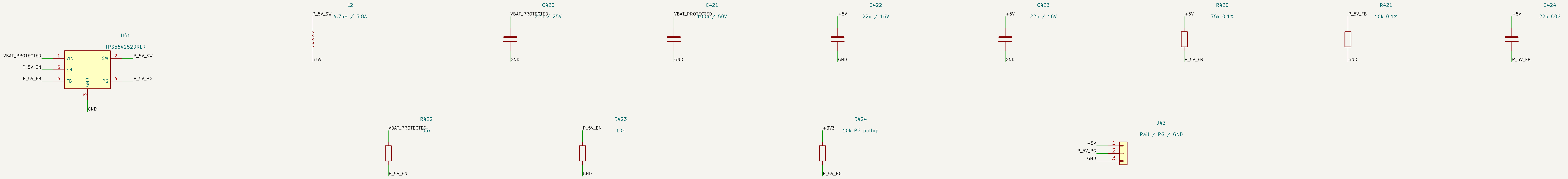
Q1 loss at 10 A: 1 W using 10 mOhm at 25 C; hotter silicon has higher resistance.
F1/holder/cable limits, transient overvoltage and thermal layout are open release checks.
J40 generic footprint is a visible placeholder; DO NOT use a small signal header for battery current.

SERVO: 6.6 V; allocation 4 A; TPS564252 (4 A device rating)


U40
TPS564252DRLR
VBAT_PROTECTED 1 VIN
P_SERVO_EN 5 EN
P_SERVO_FB 6 FB
GND 3
SW 2 P_SERVO_SW
PG 4 P_SERVO_PG
GND
L1
P_SERVO_SW
4.7uH / 5.8A
+6V6_SERVO
C410
VBAT_PROTECTED
22u / 25V
GND
C411
VBAT_PROTECTED
100n / 50V
GND
C412
+6V6_SERVO
22u / 16V
GND
C413
+6V6_SERVO
22u / 16V
GND
R410
+6V6_SERVO
100k 1%
P_SERVO_FB
R411
P_SERVO_FB
10k 1%
GND
C414
+6V6_SERVO
22p C0G
P_SERVO_FB
R412
VBAT_PROTECTED
47k
P_SERVO_EN
R413
P_SERVO_EN
10k
GND
R414
+3V3
10k PG pullup
P_SERVO_PG
J42
Rail / PG / GND
+6V6_SERVO 1
P_SERVO_PG 2
GND 3

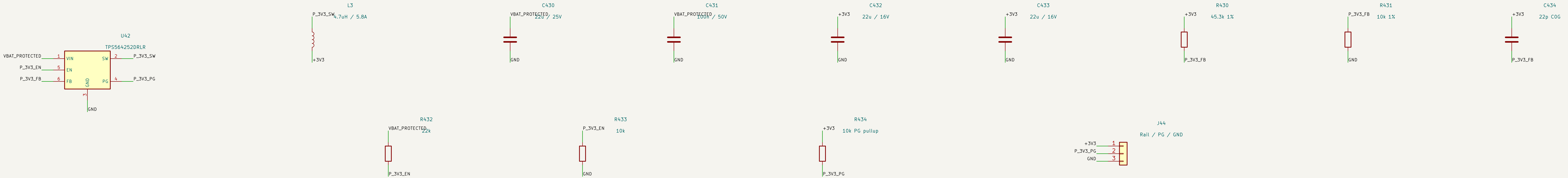
Local CIN beside VIN/GND; short SW loop. Return FB lower resistor to quiet local GND.
44uF nominal COUT; effective value/stability and full-load thermal performance need validation.

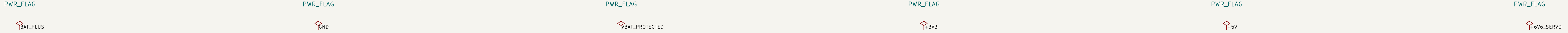
5V: 5.1 V; allocation 3.3 A; TPS564252 (4 A device rating)


U41
TPS564252DRLR
VBAT_PROTECTED 1 VIN
P_5V_EN 5 EN
P_5V_FB 6 FB
GND 3
SW 2 P_5V_SW
PG 4 P_5V_PG
GND
L2
P_5V_SW
4.7uH / 5.8A
+5V
C420
VBAT_PROTECTED
22u / 25V
GND
C421
VBAT_PROTECTED
100n / 50V
GND
C422
+5V
22u / 16V
GND
C423
+5V
22u / 16V
GND
R420
+5V
75k 0.1%
P_5V_FB
R421
P_5V_FB
10k 0.1%
GND
C424
+5V
22p C0G
P_5V_FB
R422
VBAT_PROTECTED
33k
P_5V_EN
R423
P_5V_EN
10k
GND
R424
+3V3
10k PG pullup
P_5V_PG
J43
Rail / PG / GND
+5V 1
P_5V_PG 2
GND 3

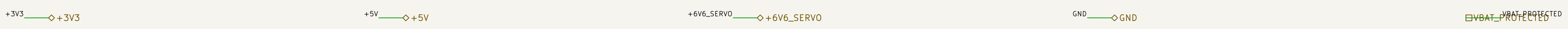
Local CIN beside VIN/GND; short SW loop. Return FB lower resistor to quiet local GND.
44uF nominal COUT; effective value/stability and full-load thermal performance need validation.

3V3: 3.318 V; allocation 0.8 A; TPS564252 (4 A device rating)


U42
TPS564252DRLR
VBAT_PROTECTED 1 VIN
P_3V3_EN 5 EN
P_3V3_FB 6 FB
GND 3
SW 2 P_3V3_SW
PG 4 P_3V3_PG
GND
L3
P_3V3_SW
4.7uH / 5.8A
+3V3
C430
VBAT_PROTECTED
22u / 25V
GND
C431
VBAT_PROTECTED
100n / 50V
GND
C432
+3V3
22u / 16V
GND
C433
+3V3
22u / 16V
GND
R430
+3V3
45.3k 1%
P_3V3_FB
R431
P_3V3_FB
10k 1%
GND
C434
+3V3
22p C0G
P_3V3_FB
R432
VBAT_PROTECTED
22k
P_3V3_EN
R433
P_3V3_EN
10k
GND
R434
+3V3
10k PG pullup
P_3V3_PG
J44
Rail / PG / GND
+3V3 1
P_3V3_PG 2
GND 3


PWR_FLAG
BAT_PLUS
PWR_FLAG
GND
PWR_FLAG
VBAT_PROTECTED
PWR_FLAG
+3V3
PWR_FLAG
+5V
PWR_FLAG
+6V6_SERVO

Local CIN beside VIN/GND; short SW loop. Return FB lower resistor to quiet local GND.
44uF nominal COUT; effective value/stability and full-load thermal performance need validation.


+3V3
+3V3
+5V
+5V
+6V6_SERVO
+6V6_SERVO
GND
GND
VBAT_PROTECTED
VBAT_PROTECTED

Engineering review; actuator/protection and mechanical sign–off pending.
PRELIMINARY – incomplete board; not for fabrication
Team 4
Sheet: /Power/
File: Power.kicad_sch
Title: Power — AA input and three buck converters
Size: A2
Date: 2026–10–23
Rev: 0.3
KiCad E.D.A. 10.0.4
Id: 5/5